

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 0606

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)
Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

SCENARIO-BASED QUESTIONS

Code of Conduct:

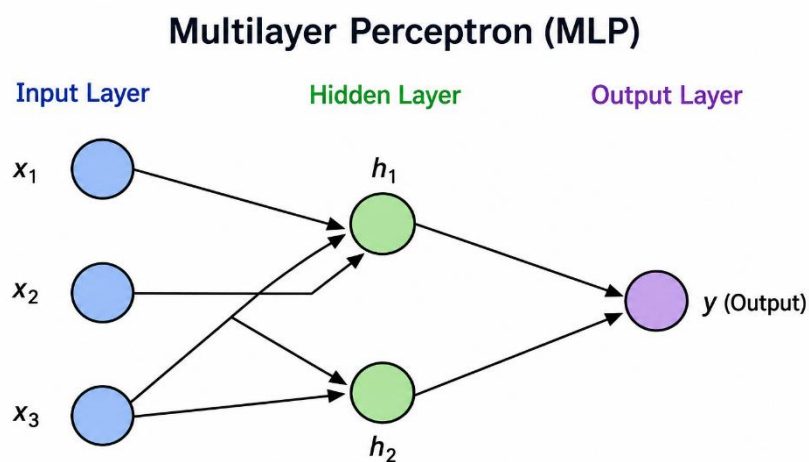
I _M. Vidhya(192412151)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

1. In the context of neural networks, consider a multilayer perceptron used for binary classification. Explain how the backpropagation algorithm updates the weights in the network during training. Illustrate the process using a simple example by describing the forward pass, error calculation, and backward propagation of errors. Discuss how learning rate and activation functions influence the convergence and accuracy of the model.

Sol)



1. Forward Pass

In the forward pass, the input values are passed from the input layer to the hidden layer and then to the output layer.

Each neuron calculates:

$$\text{Net Input} = \sum (w_i x_i) + b$$

where:

- x_i = Input
- w_i = Weight
- b = Bias

The activation function converts the net input into the output.

Example

Given:

- Input $x = 2$
- Weight $w = 0.6$
- Bias $b = 0.2$

Net Input:

$$Net = (2 \times 0.6) + 0.2 = 1.4$$

Using the Sigmoid activation function:

$$Output = \frac{1}{1 + e^{-1.4}}$$
$$Output \approx 0.80$$

Thus, the predicted output is 0.80.

2. Error Calculation

Suppose:

- Actual Output (Target) = 1
- Predicted Output = 0.80

The error is calculated using Mean Squared Error (MSE):

$$E = \frac{1}{2} (Target - Output)^2$$
$$E = \frac{1}{2} (1 - 0.80)^2$$
$$E = \frac{1}{2} (0.20)^2$$
$$E = 0.02$$

The objective is to reduce this error during training.

3. Backward Propagation

The calculated error is propagated backward through the network.

Weights are updated using the formula:

$New\ Weight = Old\ Weight - \eta \frac{\partial E}{\partial W}$
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where:

- η = Learning Rate
- $\frac{\partial E}{\partial W}$ = Gradient of error

Example

Old Weight = 0.60

Learning Rate = 0.10

Gradient = -0.20

$$\text{New Weight} = 0.60 - 0.10(-0.20)$$

$$\text{New Weight} = 0.60 + 0.02$$

$$\boxed{\text{New Weight} = 0.62}$$

The updated weight reduces the prediction error.

This process continues for every training sample until the model converges.

4. Influence of Learning Rate

The learning rate determines how much the weights change during each update.

- **Small Learning Rate (0.001):**
 - Slow convergence
 - Stable learning
 - Requires more iterations
- **Large Learning Rate (1.0):**
 - Faster learning
 - May overshoot the minimum error
 - Can make training unstable
- **Moderate Learning Rate (0.01–0.1):**
 - Faster convergence
 - Better accuracy
 - More stable training

5. Influence of Activation Functions

Activation functions introduce non-linearity into the network.

Sigmoid

- Output range: 0 to 1
- Suitable for binary classification
- Can suffer from vanishing gradient

ReLU (Rectified Linear Unit)

$$ReLU(x) = \max(0, x)$$

- Faster training
- Reduces vanishing gradient
- Widely used in deep neural networks

Tanh

- Output range: -1 to 1
- Better than Sigmoid for hidden layers
- Still affected by vanishing gradients

Working Steps of Backpropagation

1. Initialize weights randomly.
2. Perform forward propagation to calculate output.
3. Compute the prediction error.
4. Propagate the error backward.
5. Update weights using gradient descent.
6. Repeat until the error becomes minimum.

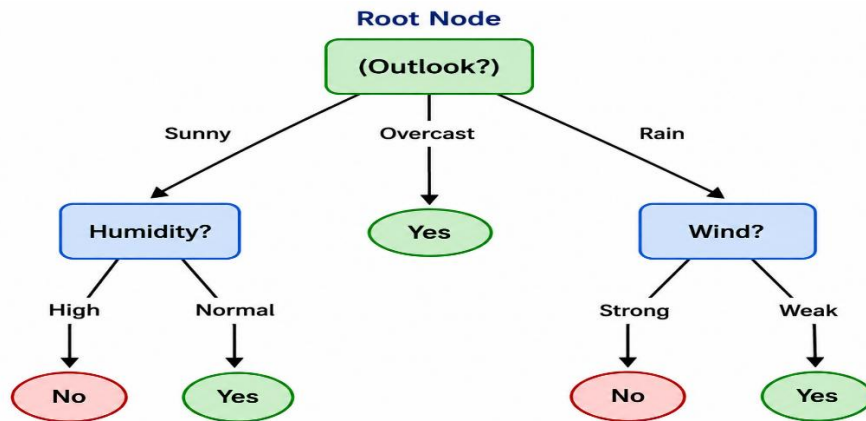
Conclusion

Backpropagation is the learning algorithm used in multilayer perceptrons. It updates the weights by propagating the error backward and minimizing the loss function through gradient descent. The choice of learning rate and activation function significantly affects the convergence speed and classification accuracy of the neural network.

2. With the help of a neat diagram, explain the architecture of a decision tree learning system. Describe how features are selected using measures such as entropy and information gain. Illustrate the process of tree construction and pruning using a small

example dataset, and explain how overfitting can be controlled in decision tree models.

Sol)



1. Architecture of a Decision Tree

A Decision Tree is a supervised learning algorithm used for classification and regression.

It consists of:

- Root Node: Represents the first feature selected for splitting.
- Decision Node: Tests a feature and divides the data.
- Leaf Node: Represents the final predicted class or output.
- Branches: Show the possible outcomes of each decision.

2. Feature Selection using Entropy and Information Gain

Entropy

Entropy measures the impurity or uncertainty of the dataset.

$$Entropy(S) = - \sum p_i \log_2 p_i$$

- Entropy = 0 → Pure dataset.
- Higher entropy → More uncertainty.

Information Gain

Information Gain measures the reduction in entropy after splitting.

$$Information\ Gain = Entropy(S) - Entropy_{after\ split}$$

The feature with the highest Information Gain is selected as the root node.

3. Example Dataset

Outlook	Humidity	Wind	Play Tennis
Sunny	High	Weak	No
Sunny	Normal	Weak	Yes
Overcast	High	Strong	Yes
Rain	High	Weak	Yes
Rain	Normal	Strong	No

- The algorithm calculates Entropy for each feature.
- Suppose Outlook has the highest Information Gain.
- Therefore, Outlook becomes the Root Node.
- The remaining features are used for further splitting until all leaf nodes contain a single class.

4. Tree Construction

1. Start with the complete dataset.
2. Calculate entropy for all features.
3. Compute Information Gain.
4. Select the feature with the highest Information Gain as the root node.
5. Split the dataset into subsets.
6. Repeat the process for each subset until all records are classified.

5. Pruning

Pruning removes unnecessary branches from the decision tree.

Types of Pruning

- Pre-Pruning: Stops tree growth early based on conditions like maximum depth or minimum samples.
- Post-Pruning: Builds the full tree first, then removes branches that do not improve accuracy.

Pruning reduces model complexity and improves generalization.

6. Controlling Overfitting

Overfitting occurs when the tree memorizes the training data and performs poorly on new data.

It can be controlled by:

- Limiting the maximum depth of the tree.
- Setting a minimum number of samples required for splitting.
- Applying pre-pruning or post-pruning.
- Using cross-validation to evaluate model performance.

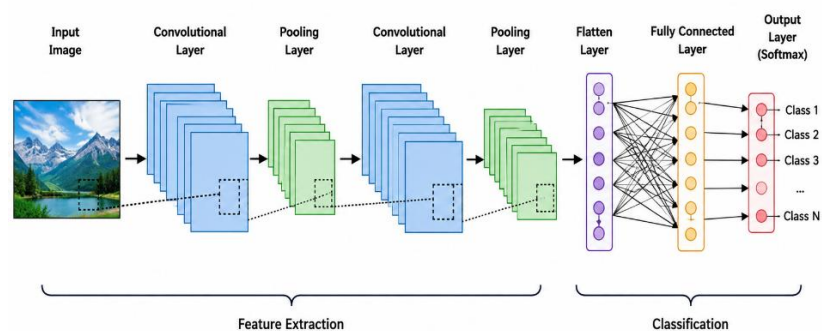
Conclusion

A Decision Tree Learning System classifies data by selecting the best features using Entropy and Information Gain. The tree is constructed by recursively splitting the data, and pruning techniques help prevent overfitting, resulting in better accuracy and improved performance on unseen data.

3. In the context of Convolutional Neural Networks (CNNs), explain the different layers involved, including convolutional, pooling, and fully connected layers. Describe how feature maps are generated and how backpropagation updates weights in CNNs. Illustrate the process with an example of image classification and discuss factors affecting model performance.

Sol)

1. Architecture of CNN



2. Layers in CNN

a) Convolutional Layer

- The convolutional layer is the first hidden layer in a CNN.
- It applies filters (kernels) to the input image.
- It extracts important features such as edges, corners, textures, and shapes.
- The output of this layer is called a feature map.

b) Pooling Layer

- The pooling layer reduces the size of the feature maps.
- It removes unnecessary information while preserving important features.
- The most common pooling method is Max Pooling.
- Pooling reduces computation and helps prevent overfitting.

c) Flatten Layer

- The flatten layer converts the 2D feature maps into a 1D vector.
- This vector is given as input to the fully connected layer.

d) Fully Connected Layer

- Every neuron is connected to all neurons in the previous layer.
- It combines the extracted features and performs the final classification.

e) Output Layer

- Produces the final prediction.
- Softmax is used for multi-class classification.
- Sigmoid is used for binary classification.

3. Feature Map Generation

- The input image is scanned using a small filter (kernel).
- The filter performs convolution by multiplying pixel values and summing them.
- An activation function such as ReLU is applied.
- The output obtained is called a feature map.
- Multiple filters generate multiple feature maps, each detecting different features.

4. Backpropagation in CNN

Backpropagation updates the weights during training.

Steps:

1. Perform forward propagation.
2. Compare predicted output with actual output.
3. Calculate the loss (error).
4. Propagate the error backward through the network.
5. Update filter weights using Gradient Descent.
6. Repeat until the loss becomes minimum.

Weight Update Formula:

$$W_{new} = W_{old} - \eta \frac{\partial L}{\partial W}$$

where:

- W = Weight
- η = Learning rate
- L = Loss function

5. Example – Image Classification

Suppose a CNN is trained to classify images into Cat, Dog, Bird, and Car.

- The convolutional layers detect edges and shapes.
- Pooling layers reduce image size.
- Fully connected layers classify the object.
- The output layer predicts the class with the highest probability.

Example Output:

- Cat $\rightarrow$ 0.92
- Dog $\rightarrow$ 0.05
- Bird $\rightarrow$ 0.02
- Car $\rightarrow$ 0.01

Hence, the image is classified as Cat.

6. Factors Affecting CNN Performance

- Quality and quantity of training data.
- Number and size of filters.

- Learning rate.
- Choice of activation function (ReLU, Sigmoid).
- Number of epochs.
- Batch size.
- Regularization techniques (Dropout).
- Computational power (GPU).

Conclusion

A Convolutional Neural Network (CNN) consists of convolutional, pooling, flatten, fully connected, and output layers. It automatically extracts image features, updates weights using backpropagation, and accurately classifies images. CNNs are widely used in image recognition, medical imaging, facial recognition, and object detection.

4. In the context of classification problems, explain how different evaluation metrics such as Accuracy, Precision, Recall, F1-score, and ROC Curve are used to assess model performance. Using a sample confusion matrix, demonstrate the calculation of these metrics and discuss their importance in applications such as medical diagnosis and fraud detection.

Sol)

Sample Confusion Matrix

	Predicted Positive	Predicted Negative
Actual Positive	TP = 40	FN = 10
Actual Negative	FP = 5	TN = 45

1. Accuracy

Accuracy measures the overall correctness of the model.

Formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Calculation:

$$\begin{aligned}
&= \frac{40 + 45}{40 + 45 + 5 + 10} \\
&= \frac{85}{100} = 0.85
\end{aligned}$$

Accuracy = 85%

2. Precision

Precision measures how many predicted positive cases are actually positive.

Formula:

$$Precision = \frac{TP}{TP + FP}$$

Calculation:

$$= \frac{40}{40 + 5} = \frac{40}{45} = 0.889$$

Precision = 88.9%

3. Recall (Sensitivity)

Recall measures how many actual positive cases are correctly identified.

Formula:

$$Recall = \frac{TP}{TP + FN}$$

Calculation:

$$= \frac{40}{40 + 10} = \frac{40}{50} = 0.80$$

Recall = 80%

4. F1-Score

F1-score is the harmonic mean of Precision and Recall.

Formula:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

Calculation:

$$= \frac{2 \times 0.889 \times 0.80}{0.889 + 0.80} = 0.842$$

F1-score = 84.2%

5. ROC Curve

- ROC (Receiver Operating Characteristic) Curve plots:
 - X-axis: False Positive Rate (FPR)
 - Y-axis: True Positive Rate (TPR)
- A model with a curve closer to the top-left corner has better performance.
- The Area Under the Curve (AUC) indicates overall classifier quality.
- Higher AUC = Better model performance.

Importance in Real-World Applications

Medical Diagnosis

- High Recall is important to detect as many disease cases as possible.
- Missing a patient with a disease (False Negative) can be dangerous.

Fraud Detection

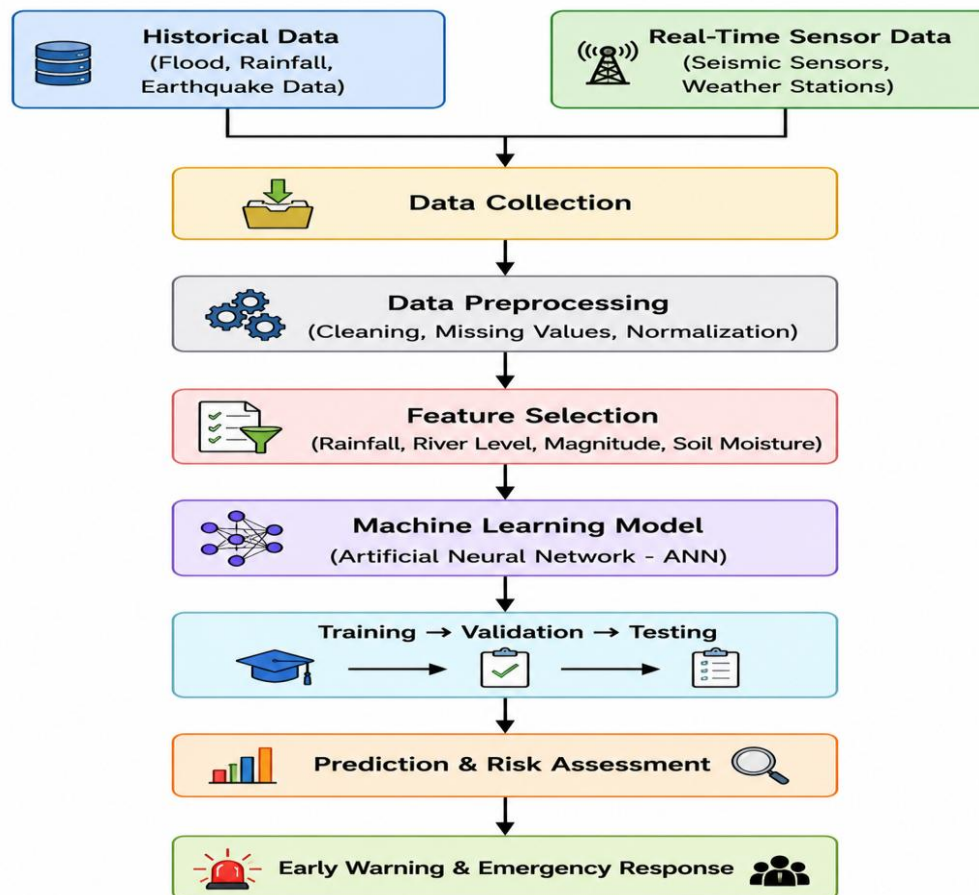
- High Precision is important to reduce false fraud alerts.
- It ensures that only genuine fraudulent transactions are flagged.

Conclusion

Evaluation metrics such as Accuracy, Precision, Recall, F1-score, and ROC Curve provide a complete assessment of a classification model. They help select the best model for real-world applications like medical diagnosis, fraud detection, spam filtering, and credit risk prediction.

5.Design a machine learning system for disaster management to predict events such as floods and earthquakes using both historical and real-time data sources. Explain the selection of an appropriate model, including whether a neural network or any other algorithm is suitable for this task, along with justification. Describe the data preprocessing steps and feature selection techniques required to improve model performance. Further, explain the complete process of training, validating, and evaluating the model, and discuss how such a system can assist authorities in early warning, risk assessment, and effective emergency response.

Sol)



1. Data Collection

The system collects data from:

- Historical Data: Previous flood records, earthquake history, rainfall, river levels.
- Real-Time Data: Weather stations, IoT sensors, satellite images, seismic sensors, GPS.

2. Selection of Machine Learning Model

Model Chosen: Artificial Neural Network (ANN)

Justification

- Learns complex relationships between multiple features.
- Handles large amounts of historical and real-time data.
- Provides high prediction accuracy.
- Suitable for nonlinear disaster prediction problems.

3. Data Preprocessing

To improve model performance:

- Remove missing values.
- Remove duplicate records.
- Normalize numerical data.
- Encode categorical values.
- Remove noisy or incorrect data.

4. Feature Selection

Important features include:

- Rainfall
- River Water Level
- Temperature
- Humidity
- Soil Moisture
- Earthquake Magnitude
- Seismic Activity
- Latitude and Longitude

Selecting only relevant features improves prediction accuracy and reduces training time.

5. Model Training, Validation, and Evaluation

Training

- The ANN is trained using historical disaster data.
- The model learns patterns from previous events.

Validation

- Validation data is used to tune model parameters and avoid overfitting.

Testing

- The trained model predicts future disasters using unseen data.

Evaluation Metrics

The model performance is measured using:

- Accuracy
- Precision
- Recall

- F1-score

Higher values indicate better prediction performance.

6. Applications in Disaster Management

The system helps authorities by:

- Providing early warning alerts before floods or earthquakes.
- Identifying high-risk areas for evacuation.
- Supporting risk assessment and disaster planning.
- Assisting emergency teams in resource allocation and rescue operations.
- Reducing loss of life and property through timely response.

Conclusion

A machine learning-based disaster management system using an Artificial Neural Network (ANN) effectively predicts floods and earthquakes by analyzing historical and real-time data. Proper preprocessing, feature selection, and model evaluation improve prediction accuracy, enabling early warning, better risk assessment, and efficient emergency response.

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